

Arithmophysics: The Rubik's Cube Continuation

Exact color moves, boundary charges, and additive energy descent

Christophe Michaels research programme

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Directive

Investigate Christophe Michaels's Rubik's Cube proposal as an operational mathematical idea: find a repeatable sequence of exact transformations that organizes coupled prime colors, preserves the authentic arithmetic source, and controls the complete energy cost of moving between scales.

The task is to derive a useful **scale-increment inequality**. Begin with the exact boundary-charge move in Section 3 and the fixed-packet increment in Section 4. Use the separated-packet theorem in Section 5 if its independently controlled part helps. A successful sequence must include every interaction and every unfinished piece it changes.

There is a concrete correction to carry forward: **additive control can close the growth argument**. A uniform constant cost per division by six gives logarithmic energy. An arbitrarily small power allowance also gives subpower energy. The previous observation of increments near 0.05 is a finite diagnostic; the next task is a uniform arithmetic estimate for their complete signed expression.

Use the cube idea to formulate moves, invariants, and quantitative costs. Execute the analytical attempt and targeted computations. Give the strongest proved inequality first, then its full return. Keep each failed estimate specific to the calculation that failed. The algebraic identities and conditional returns below are available; the arithmetic estimates explicitly marked as targets are to be proved.

1. The object that every move must preserve

For integer $N \geq 1$, define

$$M(x) = \sum_{n \leq x} \mu(n), \quad c_N(n) = \frac{\mu(n)}{n}, \quad B_N(m, n) = \min(m, n).$$

The physical charge of a coefficient vector v at n is nv_n . Vectors at smaller horizons are extended by zero when paired at a larger horizon. Write $\|v\|_B^2 = v^* B_N v$. Then

$$\begin{aligned} R(N) &= \|c_N\|_B^2 = \sum_{a, b \leq N} \frac{\mu(a)\mu(b)}{\max(a, b)} \\ &= \sum_{k=1}^{N-1} \frac{M(k)^2}{k(k+1)} + \frac{M(N)^2}{N}. \end{aligned} \tag{1}$$

Set $R(0) = 0$ and $\mathcal{S}(X) = \max_{0 \leq n \leq X} R(n)$. The terminal stays included. The elementary bound is $\mathcal{S}(X) \leq 2X$. The final target is

$$R(N) \ll_{\varepsilon} N^{\varepsilon} \quad \text{for every } \varepsilon > 0. \tag{Goal}$$

Two different scale changes will appear:

Operation	Smaller scale	Role
Fixed-ratio step	$L = \lfloor N/6 \rfloor$	Telescope additive energy increments
Sixth-root step	$N^{1/6}$	Supply a smaller-envelope factor with exponent below six

Keep these scales distinct. All noninteger summation limits are floored, and all logarithms are natural.

Optional compression, with its exact return

The existing source-independent mesh is

$$x_0 = 0, \quad x_{r+1} = \min\{N, x_r + \max(1, \lfloor N^{-1/6} x_r^{2/3} \rfloor)\}.$$

Compute its integer step as the sixth root of $\lfloor x_r^4/N \rfloor$. For $a < n \leq b$ in a mesh interval, the B_N -orthogonal boundary projection is

$$\mathcal{T}_N e_n = \frac{b-n}{b-a} e_a + \frac{n-a}{b-a} e_b, \quad e_0 = 0.$$

For $p_N = \mathcal{T}_N c_N$,

$$R(N) = E_{\mathcal{P}}(N) + E_{\text{res}}(N), \quad 0 \leq E_{\text{res}}(N) \leq C_0 = 2^{-4/3}, \quad E_{\mathcal{P}}(N) \geq \frac{M(N)^2}{N}. \quad (2)$$

The residual cap holds for real physical charges bounded by one. Consequently

$$|[E_{\mathcal{P}}(N) - E_{\mathcal{P}}(L)] - [R(N) - R(L)]| \leq C_0. \quad (3)$$

This compares the actual projections at their respective horizons; it requires no assumption that their meshes are nested.

2. Give the cube its exact arithmetic meaning

For a finite prime set $\mathcal{C}$, put $q = \prod_{p \in \mathcal{C}} p$. A parent m coprime to q generates vertices

$$m \prod_{p \in A} p, \quad A \subseteq \mathcal{C}.$$

The subset incidence is a Boolean cube with $|\mathcal{C}|$ coordinates. For $\{2, 3, 5\}$, its eight vertices are

$$m, 2m, 3m, 5m, 6m, 10m, 15m, 30m.$$

Their physical charges are $\mu(m)(-1)^{|A|}$. The Green metric couples these vertices to other packets. At a finite horizon, vertices above N form the unfinished part of the packet accounting.

Define the actual algebra of the proposed moves. Useful moves include:

1. **Regroup colors.** Change how a fixed signed source is partitioned and retain every mixed Gram term.
2. **Complete a packet.** Extract a complete divisor family and define its complement by exact subtraction from c_N .

3. **Carry a prefix to its boundary.** Preserve its total charge and record its internal orthogonal energy, as in Section 3.
4. **Separate a zero-charge block and a dilated copy.** Use the exact orthogonality and cost in Section 5.
5. **Change coordinates or compress.** Carry the transformed metric or the discarded residual so the original energy is recovered exactly.

For an invertible coordinate change $c = Uy$, the metric becomes U^*B_NU . A phase change in the colors changes the source unless its inverse or a proved return estimate is retained. The terminal parity source is always the original Möbius source.

On a fixed finite horizon,

$$\mathcal{D}_{d,N}e_m = d^{-1}e_{dm}\mathbf{1}_{\{dm \leq N\}}, \quad \mathcal{D}_{p,N}\mathcal{D}_{q,N} = \mathcal{D}_{q,N}\mathcal{D}_{p,N}. \quad (4)$$

Thus reordering the same prime factors leaves their product unchanged. Projection, changing cutoffs, or changing the decomposition can create additional terms. If order is supposed to supply a gain, calculate the relevant commutator or remainder explicitly, for example $[\mathcal{T}_N, \mathcal{D}_{p,N}]c_N$, and pay for its full signed contribution.

The desired sequence has a declared invariant and a return after each complete group of moves. It should expose an estimate for the authentic arithmetic signs. Arbitrary sign rotations selected to minimize the energy require a proved transfer back to those signs.

3. First move: cancel the unchanged prefix energy exactly

For $N \geq 6$, let $L = \lfloor N/6 \rfloor$, let $u = c_N - c_L$, and put

$$b_L = \frac{M(L)}{L}e_L, \quad r_L = c_L - b_L, \quad w_{N,L} = b_L + u.$$

The prefix residual r_L has zero physical total, is supported through L , and is B_N -orthogonal to every source supported at indices at least L . Therefore

$$R(N) = \|r_L\|_B^2 + \|w_{N,L}\|_B^2, \quad R(L) = \|r_L\|_B^2 + \frac{M(L)^2}{L}. \quad (5)$$

This is an exact move that preserves the completed prefix contribution. Its internal energy cancels from the comparison. The remaining scale increment is

$$\boxed{\Delta_6 R(N) := R(N) - R(L) = \|w_{N,L}\|_B^2 - \frac{M(L)^2}{L}.} \quad (6)$$

Two equivalent formulas provide independent checks and different analytical entry points:

$$\Delta_6 R(N) = 2M(L) \sum_{L < n \leq N} \frac{\mu(n)}{n} + \sum_{L < a, b \leq N} \frac{\mu(a)\mu(b)}{\max(a, b)}, \quad (7)$$

$$\Delta_6 R(N) = \sum_{k=L}^{N-1} \frac{M(k)^2}{k(k+1)} + \frac{M(N)^2}{N} - \frac{M(L)^2}{L}. \quad (8)$$

Attempt a joint bound for (6) or (7), retaining its inherited boundary charge. Bounding the tail square and the cross term separately is useful only when their total allowance still closes.

The carried physical charge at L is $M(L)$, which can exceed one. If compression is applied to $w_{N,L}$, insert L into the mesh and leave that boundary charge exact. Then the discarded part comes only from the bounded tail charges, so the residual cap remains available. Label this refined projection separately from the original mesh.

The primary arithmetic target

Seek, for every $\varepsilon > 0$, a constant independent of N such that

$$\boxed{\Delta_6 R(N) \leq C_\varepsilon N^\varepsilon [1 + \mathcal{S}(N^{1/6})]^\gamma, \quad 0 \leq \gamma < 6.} \quad (\text{A})$$

The case $\gamma = 0$ already closes by fixed-ratio telescoping. A uniform constant upper bound for $\Delta_6 R$ is a stronger candidate that would yield $R(N) = O(\log N)$. The fifth-power budget $\gamma = 5$ matches the existing sixth-root descent. Preserve the most flexible proved estimate rather than require a constant budget.

Equation (A) is a sufficient target, not a consequence of the preceding identities. Its proof must exploit an independent arithmetic relation in the actual signed source.

4. Second move: the fixed-packet diagonal has a bounded increment

Fix $\mathcal{C} = \{2, 3\}$, $q = 6$, and define

$$P_{23}(y) = \sum_{d|6} \mu(d) \mathbf{1}_{\{d \leq y\}}, \quad K_{23}(m, n) = \sum_{d, e|6} \frac{\mu(d)\mu(e)}{\max(dm, en)}.$$

For the completed family

$$g_N^{23} = \sum_{\substack{m \leq N/6 \\ (m, 6)=1}} \sum_{d|6} \frac{\mu(m)\mu(d)}{dm} e_{dm}, \quad t_N^{23} = c_N - g_N^{23},$$

the exact decomposition is

$$R(N) = D_{23}(N) + \mathcal{I}_{23}(N), \quad D_{23}(N) = \frac{2}{3} \sum_{\substack{m \leq N/6 \\ (m, 6)=1}} \frac{\mu(m)^2}{m}, \quad (9)$$

where

$$\begin{aligned} \mathcal{I}_{23}(N) = 2 \sum_{\substack{m < n \leq N/6 \\ (mn, 6)=1, \, n < 6m}} \mu(m)\mu(n)K_{23}(m, n) \\ + 2\langle g_N^{23}, t_N^{23} \rangle_B + \|t_N^{23}\|_B^2. \end{aligned} \quad (10)$$

The kernel vanishes for $\max(m, n) \geq 6 \min(m, n)$. Its exact logarithmic form is

$$K_{23}(m, n) = (mn)^{-1/2} k_{23}(\log(n/m)), \quad \widehat{k}_{23}(\xi) = \frac{|1 - 2^{-1/2+i\xi}|^2 |1 - 3^{-1/2+i\xi}|^2}{\xi^2 + 1/4}, \quad (11)$$

with $\widehat{k}(\xi) = \int k(u) e^{-i\xi u} du$ and support $[-\log 6, \log 6]$.

For $L = \lfloor N/6 \rfloor$, the diagonal increment has the elementary uniform bound

$$\begin{aligned} 0 \leq D_{23}(N) - D_{23}(L) &= \frac{2}{3} \sum_{\substack{\lfloor L/6 \rfloor < m \leq L \\ (m, 6)=1}} \frac{\mu(m)^2}{m} \\ &\leq \frac{2}{3} (1 + \log 6) < 1.862. \end{aligned} \quad (12)$$

Consequently

$$\Delta_6 R(N) = D_{23}(N) - D_{23}(L) + \mathcal{I}_{23}(N) - \mathcal{I}_{23}(L). \quad (13)$$

Priority calculation: derive and estimate the complete difference $\mathcal{I}_{23}(N) - \mathcal{I}_{23}(L)$, including every newly completed parent and every change in the unfinished terms. The exact combination may telescope even when separate squares do not.

A bound of the size in (A) on that remainder difference suffices, since (12) is already paid. It is unnecessary to require $\mathcal{I}_{23}(N) \leq 0$ at every future horizon. The reported negativity from 2838 through 10^7 is useful finite evidence for designing an estimate; the increment route permits positive values as well.

5. Third move: use the separated family whose cost is proved

The new *Parity Width and Separated-Packet Descent* paper supplies an independently controlled completed family. For $N \geq 216$, set

$$\ell_N = \min\{p \text{ prime} : p > \lfloor N^{5/6}/6 \rfloor\}, \quad q_N = 6\ell_N, \quad K_N = \lfloor N/q_N \rfloor.$$

In this section K_N denotes a parent cutoff, not the older maximum over spectral modes. Let

$$\begin{aligned} f_N(m) &= \frac{\mu(m)}{m} \mathbf{1}_{\{m \leq K_N, (m,6)=1\}}, & h_N &= (I - \mathcal{D}_2)(I - \mathcal{D}_3)f_N, \\ g_N &= (I - \mathcal{D}_{\ell_N})h_N, & t_N &= c_N - g_N. \end{aligned}$$

All displayed descendants fit below N . The base support is at most $6K_N$, its physical total is zero, and $\ell_N \geq 6K_N$. Thus

$$\langle h_N, \mathcal{D}_{\ell_N} h_N \rangle_B = 0, \quad G_N := \|g_N\|_B^2 = (1 + 1/\ell_N) \|h_N\|_B^2 \leq A_* \mathcal{S}(N^{1/6}), \quad (14)$$

where

$$A_* = \frac{18}{17} (17 + 12\sqrt{2})(7 + 4\sqrt{3}) = 500.981190149 \dots$$

The proof combines the exact coprime inverse

$$f c_{K,K} = \sum_{\substack{d \leq K \\ p|d \Rightarrow p \in \mathcal{C}}} \mathcal{D}_{d,K} c_K,$$

the child-prefix energy identity $\|\mathcal{D}_{d,K} c_K\|_B^2 = d^{-1} R(\lfloor K/d \rfloor)$, the resulting geometric norm bound, and zero-charge separation. In general a completed prime set has allowance

$$\|g_{\mathcal{C},N}\|_B^2 \leq \prod_{p \in \mathcal{C}} \left(\frac{1 + p^{-1/2}}{1 - p^{-1/2}} \right)^2 \mathcal{S}(N/q_{\mathcal{C}}).$$

The return is

$$R(N) = G_N + W_N, \quad W_N = 2\langle g_N, t_N \rangle_B + \|t_N\|_B^2. \quad (15)$$

For a direct two-scale formula put

$$V_N(x) = \sum_{\substack{m \leq K_N \\ (m,6)=1}} \mu(m) P_{23}(x/m).$$

Then

$$\langle g_N, c_N \rangle_B = \int_1^{6K_N} \frac{V_N(x)}{x^2} \left[M(x) - \ell_N^{-1} M(\ell_N x) \right] dx, \quad (16)$$

and the complete remainder is

$$W_N = \int_1^N \frac{[M(x) - V_N(x) + V_N(x/\ell_N)]^2}{x^2} dx + \frac{M(N)^2}{N} + 2 \int_1^{6K_N} \frac{V_N(x)}{x^2} \left[M(x) - \ell_N^{-1} M(\ell_N x) - (1 + \ell_N^{-1}) V_N(x) \right] dx. \quad (17)$$

Here $M(\ell_N x)$ is the actual step function, with its floor.

At scales where both families are defined,

$$\Delta_6 R(N) = G_N - G_L + W_N - W_L. \quad (18)$$

Try an increment estimate for $W_N - W_L$, retaining the exact completed difference when helpful. Since $G_L \geq 0$, (14) already pays its positive contribution at cost $A_* \mathcal{S}(N^{1/6})$. If the remainder increment has the right side of (A) with exponent $\gamma < 6$, the whole increment has exponent $\max(1, \gamma) < 6$, which still closes.

These families are generally not nested: the chosen prime and parent cutoff change with N . Account for those changes explicitly. A completed family with only a few dozen outputs controls that family; its complement remains in (17).

For calibration, the paper reports at $N = 4,000,000$:

$$\ell_N = 52919, \quad K_N = 12, \quad R(N) = 1.142012922 - 0.319880236 + 0.888378414 = 1.710511100.$$

These are respectively the completed square, signed cross term, and unfinished square. They are finite diagnostics to reproduce when this route is used.

6. Prove the return for the inequality actually obtained

For a fixed-ratio chain put $N_0 = N$, $N_{j+1} = \lfloor N_j/6 \rfloor$, and stop when $N_j < 6$. There are $O(\log N)$ steps and

$$R(N) = R(N_J) + \sum_{j=0}^{J-1} \Delta_6 R(N_j). \quad (19)$$

This identity requires no monotonicity of R .

If the complete increments have a uniform upper bound C , (19) gives $R(N) = O(\log N)$. If they satisfy $\Delta_6 R(n) \leq C_\varepsilon n^\varepsilon$, then

$$R(N) \leq \max_{0 \leq n < 6} R(n) + \frac{C_\varepsilon}{1 - 6^{-\varepsilon}} N^\varepsilon. \quad (20)$$

Thus additive bounds are a direct route to the target.

Under the more flexible (A), monotonicity of $\mathcal{S}$ and the same geometric sum give

$$1 + \mathcal{S}(X) \leq C'_\varepsilon X^\varepsilon [1 + \mathcal{S}(X^{1/6})]^\gamma \quad (0 \leq \gamma < 6). \quad (21)$$

Let $\rho = \limsup_{X \rightarrow \infty} \log(1 + \mathcal{S}(X))/\log X$. Since $\mathcal{S}(X) \leq 2X$, it is finite. Equation (21) implies

$$\rho \leq \varepsilon + \frac{\gamma}{6} \rho.$$

The estimate for every $\varepsilon > 0$ gives $\rho = 0$. This is the complete subpower return. Its sixth-root iteration has $O(\log \log X)$ steps; the fixed-ratio telescoping that precedes it has $O(\log X)$ steps.

An amortized estimate is also sufficient: directly bound $\sum_{j < J} [\Delta_6 R(N_j)]_+$ by the right side of (A) at the initial horizon. This permits individual expensive moves whose total positive cost is controlled.

If introducing a storage correction Q_N , specify it explicitly. For example, prove $Q_N \geq 0$ and

$$R(N) + Q_N \leq R(L) + Q_L + \mathcal{B}_N. \quad (22)$$

Then its intermediate terms telescope and the final nonnegative Q_N can be dropped. Prove the sign of Q_N and the total allowance for $\mathcal{B}_N$ independently, including the finite base values. A fitted correction with an assumed endpoint estimate leaves that estimate as an input.

7. Color alignment and phase width: supporting information

For $p_k = \mathcal{T}_N(\mu(n)^2 \mathbf{1}_{\{\omega(n)=k\}}/n)$, define

$$e_N = \sum_{k \text{ even}} p_k, \quad o_N = \sum_{k \text{ odd}} p_k, \quad \mathcal{A}_N = \|e_N\|_B^2 + \|o_N\|_B^2, \quad \delta_N = \frac{\|e_N - o_N\|_B^2}{\mathcal{A}_N}.$$

The background satisfies $\mathcal{A}_N \sim (36/\pi^4)N$, while the exact mismatch splits into amplitude and direction:

$$E_{\mathcal{P}} = (a_N - b_N)^2 + a_N b_N \left\| \frac{e_N}{a_N} - \frac{o_N}{b_N} \right\|_B^2, \quad a_N = \|e_N\|_B, \quad b_N = \|o_N\|_B. \quad (23)$$

Both pieces remain in any estimate. Use these observables to locate which part of the complete scale increment changes under a proposed move.

The new PDF also derives a local phase law. For $\mathcal{E}_N(t) = \|\sum_k e^{ikt} p_k\|_B^2$, it gives

$$\mathcal{E}_N\left(\pi + y \frac{(\log N)^2}{\sqrt{N}}\right) - \mathcal{E}_N(\pi) \longrightarrow 2y^2, \quad (24)$$

uniformly for bounded real y . This is the width of a fixed additive energy rise, with $\mathcal{E}_N(\pi) = E_{\mathcal{P}}(N)$ retained. Its derivation differentiates the parameter-uniform Selberg–Delange expansion. Reuse this width law if an angular move needs a quantified return; any such use must also control the parity baseline through an arithmetic estimate.

The asymptotic width is $(\log N)^2/\sqrt{N}$. Label finite fitted widths separately. There is no need to recompute this entire profile unless it changes the principal increment estimate.

8. Execute a bounded investigation and return a complete result

Begin by verifying (5)–(8), (12)–(13), and the summation in (19)–(21). These are the new exact connections that make this prompt more than another change of notation. Then choose the most promising signed expression and attempt its inequality analytically.

For each proposed move, record its input source, output decomposition, preserved quantity, every changed cross term, and its complete cost. A finite matrix search can suggest a certificate; derive the certificate uniformly from the actual arithmetic constraints. Any supremum over arbitrary bounded charges must survive the coherent controls below.

Use one reproducible script for the necessary diagnostics:

1. Compute $\Delta_6 R(N)$ by all three formulas (6)–(8), using exact rational checks at small horizons and stable historical sums at large ones.

2. Scan its positive peaks and chain budgets, including nonrecord horizons. Report maxima on declared ranges and on fresh ranges held out from any fit. Include horizons where a packet becomes complete or the selected dilation prime changes.
3. Evaluate the entire fixed-packet difference in (13), or the separated-family difference in (18), with every signed component. Keep their horizons and metrics identical in a comparison.
4. Use all-positive physical charges, positive squarefree charges, independent signs on squarefree support, and block-conditioned shuffles. When comparing two horizons, restrict the same control realization to both; preserve all zero positions. The actual Möbius recurrence must follow from arithmetic features stated in its proof.
5. Compute the returned growth allowance from the measured or proved per-step budget. Keep finite observations, candidate fits, and uniform theorems separately labeled. A mean increment alone does not control the largest increments or accumulated positive costs.

Use integer sieving, exact mesh rounding, prefix sums, and linear-time energy formulas. Large dense Green matrices are unnecessary. Reuse existing checkpoint data rather than repeat calculations that do not test the selected inequality.

Deliver, in this order:

1. **The move sequence and strongest proved inequality.** State the arithmetic input, quantifiers, cutoff range, constants, and proof.
2. **The complete return.** Apply (19)–(21), or the corresponding proved amortized estimate, and display the resulting growth bound. Include all terminals, residuals, and correction endpoints.
3. **The cost that matters.** Identify whether the new allowance is constant, logarithmic, subpower, or a fixed power. If a smaller-envelope factor appears, give its exponent and verify that its product with $1/6$ is below one.
4. **The reproducible computation.** Supply source, data, targeted figures, controls, and exact checks.
5. **If an arithmetic estimate remains:** display one explicit surviving signed expression and the precise inequality still needed. State which moves reduced its cost and which particular estimate lost the required control.

The intended outcome is a repeatable mathematical operation whose total effect is proved. A solved block is a contribution with an independent bound; a completed cube move includes the pieces displaced by that operation; the full solution is the returned bound on the original energy.

Source checkpoint

This directive continues *Arithmophysics: Parity Width and Separated-Packet Descent* (16 pages, October 6, 2026), especially Theorem 1.1, Lemma 3.1, equations (11)–(14), and Theorem 5.1. It also uses the preceding orthogonal boundary compression and parity-decay prompt. The identities and closing implications required for the main increment route are stated here so the work can proceed from this document alone.